fn expected_nb_mean_from_x_hi

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Proves soundness of `fn expected_nb_mean_from_x_hi` in `mod.rs`.

1 Hoare Triple

Precondition

`eta` ≥ 0 and `x` > 0.

Pseudocode

```
1  def expected_nb_mean_from_x_hi(eta: float, x: float):
2      q_hi = (-x).inf_exp()  #  propagates helper errors
3      gamma_lo = gamma_lo_from_x(x)  #  propagates helper errors
4      ensure_open_unit_interval(gamma_lo)  #  propagates helper errors
5
6      if eta == 0.0:  #
7          log_inv_gamma_lo = log_inv_gamma_lo_from_x(x)  # propagates helper errors
8          denom_lo = gamma_lo.neg_inf_mul(log_inv_gamma_lo)  # propagates helper errors
9          if denom_lo <= 0.0:
10             return float("inf")
11         return q_hi.inf_div(denom_lo)  # propagates helper errors
12
13     gamma_eta_hi = gamma_pow_eta_hi_from_x(eta, x)  #  propagates helper errors
14     one_minus_gamma_eta_lo = 1.0.neg_inf_sub(gamma_eta_hi)  # propagates helper errors
15     if one_minus_gamma_eta_lo <= 0.0:
16         return float("inf")
17     denom_lo = gamma_lo.neg_inf_mul(one_minus_gamma_eta_lo)  # propagates helper errors
18     if denom_lo <= 0.0:
19         return float("inf")
20     num_hi = eta.inf_mul(q_hi)  # propagates helper errors
21     return num_hi.inf_div(denom_lo)  # propagates helper errors
```

Postcondition

Theorem 1.1. For any inputs satisfying the precondition, `expected_nb_mean_from_x_hi` either returns `Err(e)`, or returns `Ok(out)` where `out` is an upper enclosure of the Definition 1 expected repetition count.

2 Proof

Assume the precondition, let $q = e^{-x}$ and $\gamma = 1 - e^{-x}$. Since $x > 0$, we have $q \in (0, 1)$ and $\gamma \in (0, 1)$.

Lemma 2.1. `expected_nb_mean_from_x_hi` only returns `Err(e)` if one of the fallible calls on lines 2, 3, 4, 13, or the unlabeled fallible arithmetic calls in the taken branch returns `Err(e)`.

Proof. Every branch return is either an explicit `float("inf")` or the result of propagating one of the listed fallible helper calls. □

Lemma 2.2. If `eta = 0` and `expected_nb_mean_from_x_hi` returns `Ok(out)`, then

$$\texttt{out} \geq \frac{e^{-x}}{\gamma \log(1/\gamma)}.$$

Proof. By the proof definitions of `inf_exp` and `gamma_lo_from_x`,

$$\texttt{q_hi} \geq q, \qquad \texttt{gamma_lo} \leq \gamma.$$

If line 4 succeeds, then `gamma_lo` lies in $(0, 1)$. By the proof of `log_inv_gamma_lo_from_x`,

$$\texttt{log_inv_gamma_lo} \leq \log(1/\gamma).$$

Therefore the rounded multiplication on line 6 computes a lower bound on the true denominator, so

$$\texttt{denom_lo} \leq \gamma \log(1/\gamma).$$

If `denom_lo` ≤ 0, the function returns $+\infty$, which is an upper bound. Otherwise `denom_lo` is a positive lower bound on the true denominator, so upward-rounded division yields

$$\texttt{out} \geq \frac{q}{\gamma \log(1/\gamma)} = \frac{e^{-x}}{\gamma \log(1/\gamma)}.$$

□

Lemma 2.3. If `eta > 0` and `expected_nb_mean_from_x_hi` returns `Ok(out)`, then

$$\texttt{out} \geq \frac{\texttt{eta} \cdot e^{-x}}{\gamma(1 - \gamma^{\texttt{eta}})}.$$

Proof. As above, `q_hi` $\geq q$ and `gamma_lo` $\leq \gamma$, with `gamma_lo` in $(0, 1)$ after line 4. By the proof of `gamma_pow_eta_hi_from_x`,

$$\texttt{gamma_eta_hi} \geq \gamma^{\texttt{eta}}.$$

Therefore

$$\texttt{one_minus_gamma_eta_lo} \leq 1 - \gamma^{\texttt{eta}}.$$

If `one_minus_gamma_eta_lo` ≤ 0 or `denom_lo` ≤ 0, the function returns $+\infty$, which is an upper bound. Otherwise both lower bounds are positive, so

$$\texttt{denom_lo} \leq \gamma(1 - \gamma^{\texttt{eta}}) \quad \text{and} \quad \texttt{num_hi} \geq \texttt{eta} \cdot e^{-x}.$$

Upward-rounded division then gives the claimed upper bound. □

Proof of Theorem 1.1. Holds by Lemma 2.1, Lemma 2.2, and Lemma 2.3. □